MID-TERMEXAMINATION
(CBCS)
B. Tech (IT/MAE/DMAM), 2nd Semester
(June, 2022)

Subject Code: BAS-106	Subject: Environmental Sciences

Time : 1 ½ Hours | Maximum Marks : 30

Note:Q. 1 is compulsory. Attempt any two questions from the rest.

Q1			**(2.5*4)**
	(a)	What devices/methods are being used by industries to control particulate emissions? Discuss any two briefly.	
	(b)	Is Gross Calorific Value and Net Calorific Value for a fuel different? When can both values for a fuel be same? Discuss.	
	(c)	Write two chemical reactionswhere atom economy of more than 80% is recorded.Calculate the atom economy of production of CO_2 by burning propane in the presence of sufficient amount of oxygen.	
	(d)	What are the causes and consequences of deforestation?	
Q2			**(5,5)**
	(a)	What are the characteristics and the type of waste water coming from a food industry? What possible treatments would you suggest to treat the waste water?	
	(b)	Discuss the following: I. Environmental effects of extracting mineral resources II. Sources and sinks of CO emissions	
Q3			**(5,5)**
	(a)	Can vegetable oil/animal fat be a potential feedstock for production of a clean fuel? Discuss. Also explain thetwo principles of green chemistry, "Prevention of hazardous chemical Synthesis" and "Designing Safer Chemicals" giving suitable examples.	
	(b)	What are hazardous wastes? Explain the methods of disposal of nuclear waste.	
Q4			**(5,5)**
	(a)	Calculate the GCV and NCV of fuel having carbon=85%, hydrogen= 8%, sulphur= 1%, nitrogen= 2% and ash= 4%. Latent heat of steam=587 cal/g.	
	(b)	What are the measures by which carbon dioxide can be sequestered from the atmosphere? Discuss in detail.	